

STA 131A: Introduction to Probability Theory

Lecture 21: Moment Generating Functions

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Announcements

Midterm 2

- Solutions and scores are posted online
- Please review the solution carefully and identify topics to revisit
- You may review your graded exam in discussion section tomorrow
- Regrade requests, if any, should follow the instructions posted on Canvas

Homework 6 is posted

- This includes re-working Midterm 2 problems and reflecting on what you might have missed and what you have reviewed
- Start early and post questions on Piazza as needed

Office hours today

- 2:30–3:30 PM at MSB 4220

Agenda

Last time:

- Covariance and correlation
- Variance of sums
- Conditional variance

Today:

- **Variance decomposition:** finishing the story of conditional variance
 - Law of total variance
 - Estimation/regression interpretation
- **Moment generating functions:** packaging moments into one function
 - Definition and examples
 - Computing moments from MGFs
 - Identifying distributions from MGFs

Recap: Conditional variance

Definition

The **conditional variance** of X given $Y = y$ is

$$\text{Var}(X \mid Y = y) = \mathbb{E} \left[(X - \mathbb{E}[X \mid Y = y])^2 \mid Y = y \right].$$

Equivalently,

$$\text{Var}(X \mid Y = y) = \mathbb{E}[X^2 \mid Y = y] - (\mathbb{E}[X \mid Y = y])^2.$$

- $\text{Var}(X \mid Y = y)$ quantifies the uncertainty in X conditioned on $Y = y$
- $\text{Var}(X \mid Y)$ is a random variable that takes value $\text{Var}(X \mid Y = y)$ when $Y = y$

Question: How does the “total uncertainty” $\text{Var}(X)$ relate to the collection of conditional variances $\text{Var}(X \mid Y = y)$?

Law of total variance

Theorem (Law of total variance)

$$\text{Var}(X) = \mathbb{E}[\text{Var}(X \mid Y)] + \text{Var}(\mathbb{E}[X \mid Y]).$$

- $\mathbb{E}[\text{Var}(X \mid Y)]$: average uncertainty that remains after observing Y .
- $\text{Var}(\mathbb{E}[X \mid Y])$: variability explained by how the conditional mean changes with Y .

Interpretation:

total variability = average within-condition variability + between-condition variability.

This is the variance analogue of the law of total expectation:

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]].$$

Proof: Why the law of total variance holds

Theorem (Law of total variance)

$$\text{Var}(X) = \mathbb{E}[\text{Var}(X \mid Y)] + \text{Var}(\mathbb{E}[X \mid Y]).$$

Proof. Let $m(Y) = \mathbb{E}[X \mid Y]$. Then

$$\text{Var}(X \mid Y) = \mathbb{E}[X^2 \mid Y] - m(Y)^2.$$

Taking expectations and using the law of total expectation,

$$\mathbb{E}[\text{Var}(X \mid Y)] = \mathbb{E}[\mathbb{E}[X^2 \mid Y]] - \mathbb{E}[m(Y)^2] = \mathbb{E}[X^2] - \mathbb{E}[m(Y)^2]. \quad (1)$$

Also, since $\mathbb{E}[m(Y)] = \mathbb{E}[X]$ by total expectation,

$$\text{Var}(\mathbb{E}[X \mid Y]) = \text{Var}(m(Y)) = \mathbb{E}[m(Y)^2] - (\mathbb{E}[m(Y)])^2 = \mathbb{E}[m(Y)^2] - (\mathbb{E}[X])^2. \quad (2)$$

Adding the two equations (1) and (2) gives

$$\mathbb{E}[\text{Var}(X \mid Y)] + \text{Var}(\mathbb{E}[X \mid Y]) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \text{Var}(X).$$

Example: Total variance in the two-group model

Example

Recall the two-group model:

$$Y \sim \text{Bernoulli}(1/2), \quad X \mid Y = 0 \sim N(0, 1), \quad X \mid Y = 1 \sim N(2, 4).$$

Thus,

$$\mathbb{E}[X \mid Y] = 2Y, \quad \text{Var}(X \mid Y) = 1 + 3Y.$$

Since $Y \sim \text{Bernoulli}(1/2)$,

$$\mathbb{E}[\text{Var}(X \mid Y)] = \mathbb{E}[1 + 3Y] = 1 + 3 \cdot \frac{1}{2} = \frac{5}{2}.$$

Also,

$$\text{Var}(\mathbb{E}[X \mid Y]) = \text{Var}(2Y) = 4\text{Var}(Y) = 4 \cdot \frac{1}{4} = 1.$$

Therefore, by the law of total variance,

$$\text{Var}(X) = \frac{5}{2} + 1 = \frac{7}{2}.$$

Estimation viewpoint: Resolved vs. remaining variability

Inverse-problem setting: Suppose X is an unknown quantity of interest, and Y is an observed measurement/data. Before observing Y , uncertainty about X is measured by $\text{Var}(X)$.

Estimation of X given Y : After observing Y , a natural prediction of X is the conditional mean

$$\hat{X} = \mathbb{E}[X \mid Y].$$

Define the residual $R = X - \hat{X}$. Then

$$\mathbb{E}[R \mid Y] = \mathbb{E}[X - \mathbb{E}[X \mid Y] \mid Y] = 0.$$

Thus, after using the information in Y , the residual has conditional mean zero.

Moreover, since $\mathbb{E}[R] = 0$, $\text{Var}(R) = \mathbb{E}[R^2]$. Therefore,

$$\text{Var}(\hat{X}) = \text{Var}(\mathbb{E}[X \mid Y]) \quad \text{and} \quad \text{Var}(R) = \mathbb{E}[R^2] = \mathbb{E}[\text{Var}(X \mid Y)].$$

Therefore, the law of total variance can be read as

$$\text{Var}(X) = \underbrace{\text{Var}(\hat{X})}_{\text{uncertainty resolved/explained by } Y} + \underbrace{\text{Var}(R)}_{\text{average posterior uncertainty left after } Y}.$$

Message: Observing Y changes the conditional mean $\mathbb{E}[X \mid Y]$. The variability of these conditional means is the explained part; the residual term is the average uncertainty left after observing Y .

Regression viewpoint: Explained vs. residual variability

Regression setting: Suppose now that

$$Y = \text{response}, \quad X = \text{predictor}.$$

The conditional mean

$$m(X) = \mathbb{E}[Y \mid X]$$

is the regression function: it gives the mean response after seeing the predictor value.

The law of total variance becomes

$$\text{Var}(Y) = \underbrace{\text{Var}(m(X))}_{\text{variation explained by the predictor}} + \underbrace{\mathbb{E}[\text{Var}(Y \mid X)]}_{\text{average within-predictor noise}}.$$

Interpretation:

- If X is informative, then $m(X)$ changes substantially with X , so $\text{Var}(m(X))$ is large.
- If responses are tightly concentrated around $m(X)$, then $\mathbb{E}[\text{Var}(Y \mid X)]$ is small.

This is the probability-theory version of the familiar idea:

$$\text{total variation} = \text{explained variation} + \text{residual variation}.$$

Pop-up quiz: Total variance

Suppose that

$$\text{Var}(X \mid Y) = 2 \quad \text{for every value of } Y, \quad \mathbb{E}[X \mid Y] = 3Y, \quad \text{Var}(Y) = 4.$$

Question: What is $\text{Var}(X)$?

- A) 2
- B) 6
- C) 14
- D) 38

Answer: D.

By the law of total variance,

$$\text{Var}(X) = \mathbb{E}[\text{Var}(X \mid Y)] + \text{Var}(\mathbb{E}[X \mid Y]) = 2 + \text{Var}(3Y) = 2 + 9 \cdot 4 = 38.$$

Follow-up: In

$$\text{Var}(X) = \mathbb{E}[\text{Var}(X \mid Y)] + \text{Var}(\mathbb{E}[X \mid Y]),$$

which term becomes small if observing Y makes X almost deterministic?

Motivation for moment generating functions

We have computed moments directly:

$$\mathbb{E}[X], \quad \mathbb{E}[X^2], \quad \text{Var}(X).$$

But repeating direct integration or summation for each can be tedious.

A **moment generating function** packages all moments into a single function:

$$M_X(s) = \mathbb{E}[e^{sX}].$$

Why is this useful?

- Differentiating M_X gives moments
- MGFs can identify distributions
- They can make sums of independent random variables easier; we will use this next lecture

Moment generating function

Definition (Moment generating function)

The **moment generating function** (or **MGF**) of a random variable X is

$$M_X(s) = \mathbb{E}[e^{sX}],$$

for values of $s \in \mathbb{R}$ where this expectation is finite.

- Discrete case:

$$M_X(s) = \sum_x e^{sx} p_X(x).$$

- Continuous case:

$$M_X(s) = \int_{-\infty}^{\infty} e^{sx} f_X(x) dx.$$

The MGF is a function of the parameter s , not a single number.

- $M_X(0) = 1$ always.
- To be useful as an MGF, $M_X(s)$ should be finite on an open interval around 0.
- Some random variables do not have a finite MGF near 0.

Examples: Bernoulli and Poisson MGFs

Example (Bernoulli MGF)

Let $X \sim \text{Bernoulli}(p)$. Then

$$P(X = 1) = p, \quad P(X = 0) = 1 - p.$$

Thus,

$$M_X(s) = \mathbb{E}[e^{sX}] = (1 - p)e^{s \cdot 0} + pe^{s \cdot 1} = 1 - p + pe^s.$$

Example (Poisson MGF)

Let $X \sim \text{Poisson}(\lambda)$. Then

$$\begin{aligned} M_X(s) &= \mathbb{E}[e^{sX}] = \sum_{k=0}^{\infty} e^{sk} e^{-\lambda} \frac{\lambda^k}{k!} \\ &= e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda e^s)^k}{k!} = e^{-\lambda} e^{\lambda e^s} = \exp[\lambda(e^s - 1)]. \end{aligned}$$

Example: Exponential MGF

Example

Let $X \sim \text{Exponential}(\lambda)$, so

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0.$$

For $s < \lambda$,

$$\begin{aligned} M_X(s) &= \int_0^{\infty} e^{sx} \lambda e^{-\lambda x} dx \\ &= \lambda \int_0^{\infty} e^{-(\lambda-s)x} dx \\ &= \frac{\lambda}{\lambda - s}. \end{aligned}$$

For $s \geq \lambda$, the integral diverges. Thus the MGF for $\text{Exponential}(\lambda)$ exists only for $s < \lambda$.

MGFs generate moments

From MGFs to moments

When the MGF exists in an interval around 0, differentiating under the expectation gives

$$M_X^{(k)}(0) = \left. \frac{d^k}{ds^k} M_X(s) \right|_{s=0} = \mathbb{E}[X^k].$$

In particular,

$$M'_X(s) = \mathbb{E}[Xe^{sX}], \quad M''_X(s) = \mathbb{E}[X^2 e^{sX}].$$

Setting $s = 0$,

$$M'_X(0) = \mathbb{E}[X], \quad M''_X(0) = \mathbb{E}[X^2].$$

Therefore,

$$\text{Var}(X) = M''_X(0) - (M'_X(0))^2.$$

Takeaway: With MGF, all moments can be computed via differentiation.

Example: Moments from Bernoulli MGFs

Example (Bernoulli)

For $X \sim \text{Bernoulli}(p)$,

$$M_X(s) = 1 - p + pe^s.$$

Thus,

$$M'_X(s) = pe^s, \quad M''_X(s) = pe^s.$$

Therefore,

$$\mathbb{E}[X] = M'_X(0) = p$$

and

$$\begin{aligned} \text{Var}(X) &= M''_X(0) - (M'_X(0))^2 \\ &= p - p^2 \\ &= p(1 - p). \end{aligned}$$

Example: Moments from Poisson MGFs

Example (Poisson)

For $X \sim \text{Poisson}(\lambda)$,

$$M_X(s) = \exp\{\lambda(e^s - 1)\}.$$

Differentiating,

$$M'_X(s) = \lambda e^s M_X(s),$$

and

$$M''_X(s) = (\lambda e^s + \lambda^2 e^{2s}) M_X(s).$$

Thus,

$$M'_X(0) = \lambda, \quad M''_X(0) = \lambda + \lambda^2.$$

Hence

$$\mathbb{E}[X] = \lambda, \quad \text{Var}(X) = \lambda.$$

Example: Moments from the exponential MGF

Example

If $X \sim \text{Exponential}(\lambda)$, then

$$M_X(s) = \frac{\lambda}{\lambda - s}, \quad s < \lambda.$$

Differentiate:

$$M'_X(s) = \frac{\lambda}{(\lambda - s)^2}, \quad M''_X(s) = \frac{2\lambda}{(\lambda - s)^3}.$$

Thus,

$$\mathbb{E}[X] = M'_X(0) = \frac{1}{\lambda}, \quad \mathbb{E}[X^2] = M''_X(0) = \frac{2}{\lambda^2}.$$

Therefore,

$$\text{Var}(X) = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}.$$

Pop-up quiz: Moments from an MGF

Suppose a random variable X has MGF

$$M_X(s) = \exp(2s + 3s^2).$$

Question: What are $\mathbb{E}[X]$ and $\text{Var}(X)$?

- A) $\mathbb{E}[X] = 2, \quad \text{Var}(X) = 3$
- B) $\mathbb{E}[X] = 2, \quad \text{Var}(X) = 6$
- C) $\mathbb{E}[X] = 5, \quad \text{Var}(X) = 3$
- D) $\mathbb{E}[X] = 5, \quad \text{Var}(X) = 6$

Answer: B. We have

$$M'_X(s) = (2 + 6s)e^{2s+3s^2}, \quad \text{and thus,} \quad M'_X(0) = 2.$$

Similarly,

$$M''_X(s) = [6 + (2 + 6s)^2]e^{2s+3s^2}, \quad \text{and thus,} \quad M''_X(0) = 10.$$

Therefore,

$$\text{Var}(X) = M''_X(0) - (M'_X(0))^2 = 10 - 4 = 6.$$

MGFs can identify distributions

Inversion property

If $M_X(s)$ is finite for all s in some open interval $(-a, a)$ around 0, then the MGF M_X uniquely determines the distribution (CDF) of X .

If two random variables have the same MGF on an open interval around $s = 0$, then they have the same distribution.

Important remarks:

- The uniqueness statement requires the MGF to exist near 0.
- There exist explicit formulas that allow us to recover the PMF or PDF (inverse transform)
- We usually identify distributions by “pattern matching” to known MGFs.

MGFs of independent sums

If X and Y are independent, then

$$\begin{aligned}M_{X+Y}(s) &= \mathbb{E}[e^{s(X+Y)}] \\&= \mathbb{E}[e^{sX} e^{sY}] \\&= \mathbb{E}[e^{sX}] \mathbb{E}[e^{sY}] \\&= M_X(s) M_Y(s).\end{aligned}$$

This is the transform version of convolution:

convolution of distributions $\longleftrightarrow$ multiplication of MGFs.

We will use this next time to analyze sums and random sums.

Wrap-up

Law of total variance

$$\text{Var}(X) = \mathbb{E}[\text{Var}(X \mid Y)] + \text{Var}(\mathbb{E}[X \mid Y]).$$

- Average remaining uncertainty plus variability explained by Y

Moment generating functions

$$M_X(s) = \mathbb{E}[e^{sX}].$$

- If M_X exists near 0, then $M_X^{(k)}(0) = \mathbb{E}[X^k]$
- $\text{Var}(X) = M_X''(0) - (M_X'(0))^2$
- MGFs can identify distributions when they exist on an interval around 0

Next lecture: MGFs for sums of independent random variables and random sums

Suggested reading: [BT08, Ch. 4.3 & 4.4]

References



Dimitri Bertsekas and John N Tsitsiklis.

Introduction to probability, volume 1.

Athena Scientific, 2nd edition, 2008.